

Helping the World

As an Industrial Engineer, Dan worked as part of a team designing a process for making interlocking blocks to build houses, schools, and hospitals in rural Thailand and Laos. Some of these areas were so poor that young people didn't stay in the region to help boost the economy and infrastructure. They preferred instead to travel to Bangkok, Bali, or Saigon to work in the international tourism industry. The new block-making equipment and methods made it easier and attractive for young, able workers to earn a good living locally and help build up their communities. The engineers on his team tested different material combinations for block strength and perfected the pressing methods and inspection techniques so each block came out the same size and consistency.

Dan accompanied students from the Cal Poly Engineers Without Borders chapter to work on a project in Thailand to design and implement a water-filtering system based on a tank full of sand and microorganisms set on a hill. Water pumped through the tank became safe to drink and distribute to a village below. The filter system was also complemented with local, home filters that worked with rainwater catches to utilize the large, annual rainfall in the region. These ensured sufficient clean water for cooking, cleaning, and drinking. Contaminated water had been a source for recurrent illness in the area for decades; the new system has worked well to improve the lives of the people there. Engineers are trained to look at problems differently, so they are valuable at solving problems in almost any area that needs improvement.

Engineers Changing the World



Dan Waldorf
Professor

"The best part of being an engineer is that you get to do something different every day. There is a new challenge, a new opportunity, or a new technological development every time you solve any given problem. It means the job is never boring, and it means that others look to you with great respect for your ability to tackle an infinite variety of problems."

